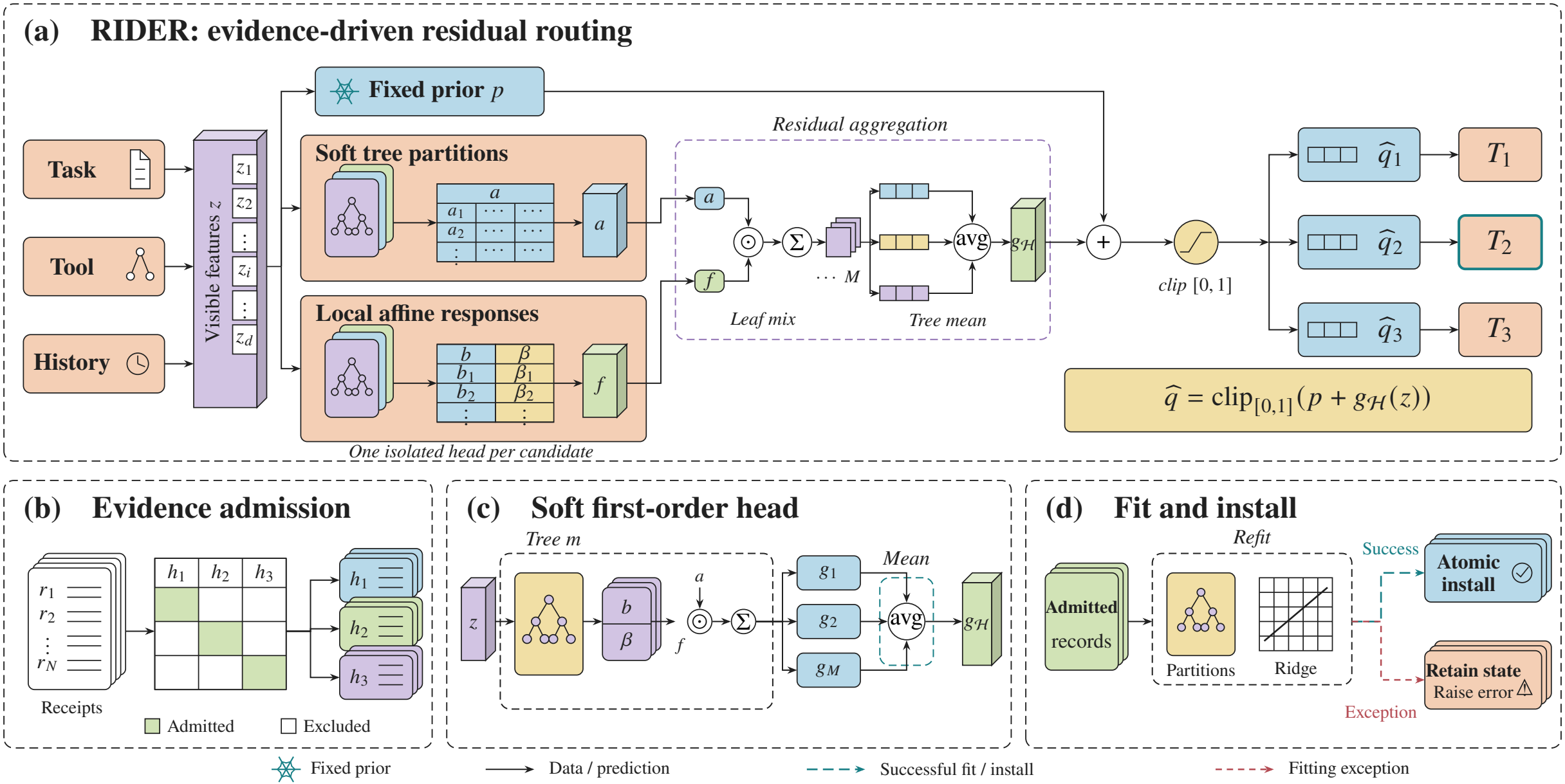

(a) RIDER: evidence-driven residual routing
Fixed prior p
Task
Tool
History
Visible features z
z_1
z_2
z_i
z_d
Soft tree partitions
a
a_1
a_2
Local affine responses
b
b_1
b_2
β
β_1
β_2
f
One isolated head per candidate
Residual aggregation
Leaf mix
Tree mean
avg
M
$g_{\mathcal{H}}$
clip $[0, 1]$
$\widehat{q}_1$
$\widehat{q}_2$
$\widehat{q}_3$
T_1
T_2
T_3
$\widehat{q} = \mathrm{clip}_{[0,1]}(p + g_{\mathcal{H}}(z))$
(b) Evidence admission
r_1
r_2
r_N
Receipts
h_1
h_2
h_3
Admitted
Excluded
(c) Soft first-order head
Tree m
z
g_1
g_2
g_M
Mean
(d) Fit and install
Refit
Admitted records
Partitions
Ridge
Success
Atomic install
Retain state
Raise error
Exception
Fixed prior
Data / prediction
Successful fit / install
Fitting exception